

Task Horizons in Real-World Usage

Success vs. task duration by platform

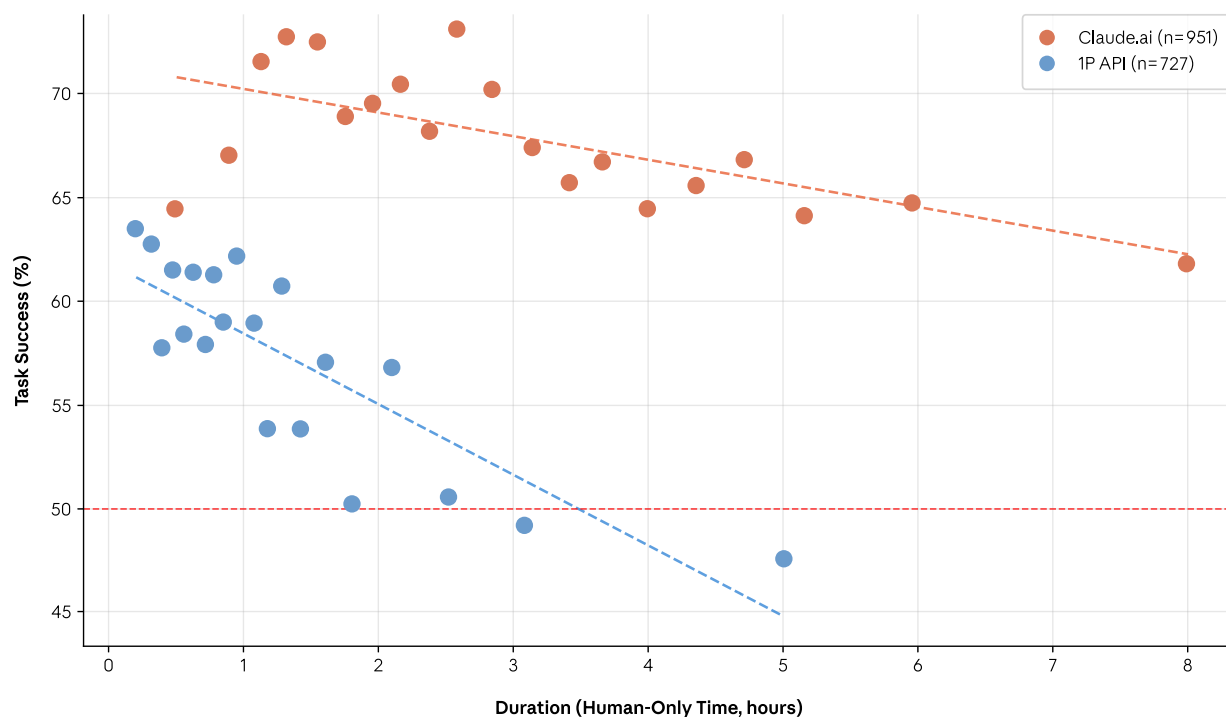


Figure 4.3: Task success vs. human-only time.

The plot shows a binned scatterplot of the bivariate relationship between task success (%) and the time the task would require a human to complete alone, all measured at the O*NET task level and split by platform. The dashed lines show the fit from a linear regression.

Recent work on AI “task horizons” ([Kwa et al., 2025](#)) finds that AI success rates decline with task duration: longer tasks are harder for models to complete. With each successive model generation, however, this decline has become shallower as models succeed on increasingly long tasks. METR operationalizes task horizon primarily as the maximum duration at which a model achieves at least 50% success, and growth in this metric has become a key indicator of AI progress.

Figure 4.3 shows a similar measure using our primitives. The plot shows task-level success rates against the human time required, all at the O*NET task level. In the API data, success rates drop from around 60% for sub-hour tasks to roughly 45% for tasks estimated to take humans 5+ hours. The fitted line

crosses the horizontal 50% success line at 3.5 hours, suggesting that API calls attain a 50% success rate for tasks that are 3.5 hours. The analogous time estimate in METR's software engineering benchmark is 2 hours for Sonnet 4.5 and about 5 hours for Opus 4.5. (The data in this report predates the release of Opus 4.5.)

Claude.ai data tells a different story. Success rates decline far slower as a function of task length. Extrapolating using the linear fit, Claude.ai would hit a 50% success rate at about 19 hours. This may reflect how multi-turn conversation effectively breaks complex tasks into smaller steps, with each turn providing a feedback loop that allows users to correct course.

It's worth noting that a fundamental difference from the METR setting is selection. METR constructs a benchmark where a fixed set of tasks is assigned to models. In our data, users choose which tasks to bring to Claude. This means observed success rates reflect not just model capability but also user judgment about what will work, the cost of setting up the problem for Claude, and the expected time savings if the task succeeds.

If users avoid tasks they expect to fail, for example, observed success rates will overstate true capability on the full distribution of potential tasks. This selection likely operates on both platforms, but in different ways: API customers select for tasks amenable to automation, while Claude.ai users select for tasks that could benefit from iteration. Also due to this selection effect, there's no guarantee that more performant models would show improvement in this plot, because users may respond to new models by providing more challenging presentations of otherwise similar O*NET tasks.

Controlled benchmarks like METR's measure the frontier of autonomous capability. Our real-world data can measure the *effective* task horizon, reflecting a mix of model capabilities and user behavior, and expanding beyond coding tasks. Both approaches find that AI can be effective for tasks requiring hours of human work.

Revisiting occupation penetration with effective AI coverage

Our [earlier work](#) found that 36% of jobs had AI usage for at least a quarter of their tasks, with about 4% reaching 75% task coverage. This measure was based only on the appearance of a task in our data, however. The primitives introduced in this report can help better characterize how AI is changing the work content of occupations.³

First, we find that task coverage is increasing. Combining across reports, 49% of jobs have seen AI usage for at least a quarter of their tasks. But incorporating that task's share of the job, and Claude's average success rate, suggests a different set of affected occupations.

We define effective AI coverage as the percent of a worker's day that can be performed successfully by Claude. It's calculated as the weighted sum of task success rates, where each task's weight is its share of the worker's time adjusted by how frequently the task occurs. The success rate comes from our primitives, the hours estimate from our previous work on productivity effects, and the frequency estimate from O*NET data, where surveyed workers indicate how often they perform the task.

The plot below shows how the effective AI coverage (y-axis) differs from task coverage alone (x-axis). The two are highly correlated, but with key differences. On the right side of the plot, occupations with high coverage—where almost all tasks appear with some frequency in Claude data—generally fall below the 45-degree line. This suggests that even 90% task coverage does not necessarily indicate large job impacts, since Claude may fail on key covered tasks or miss the most time-intensive ones.

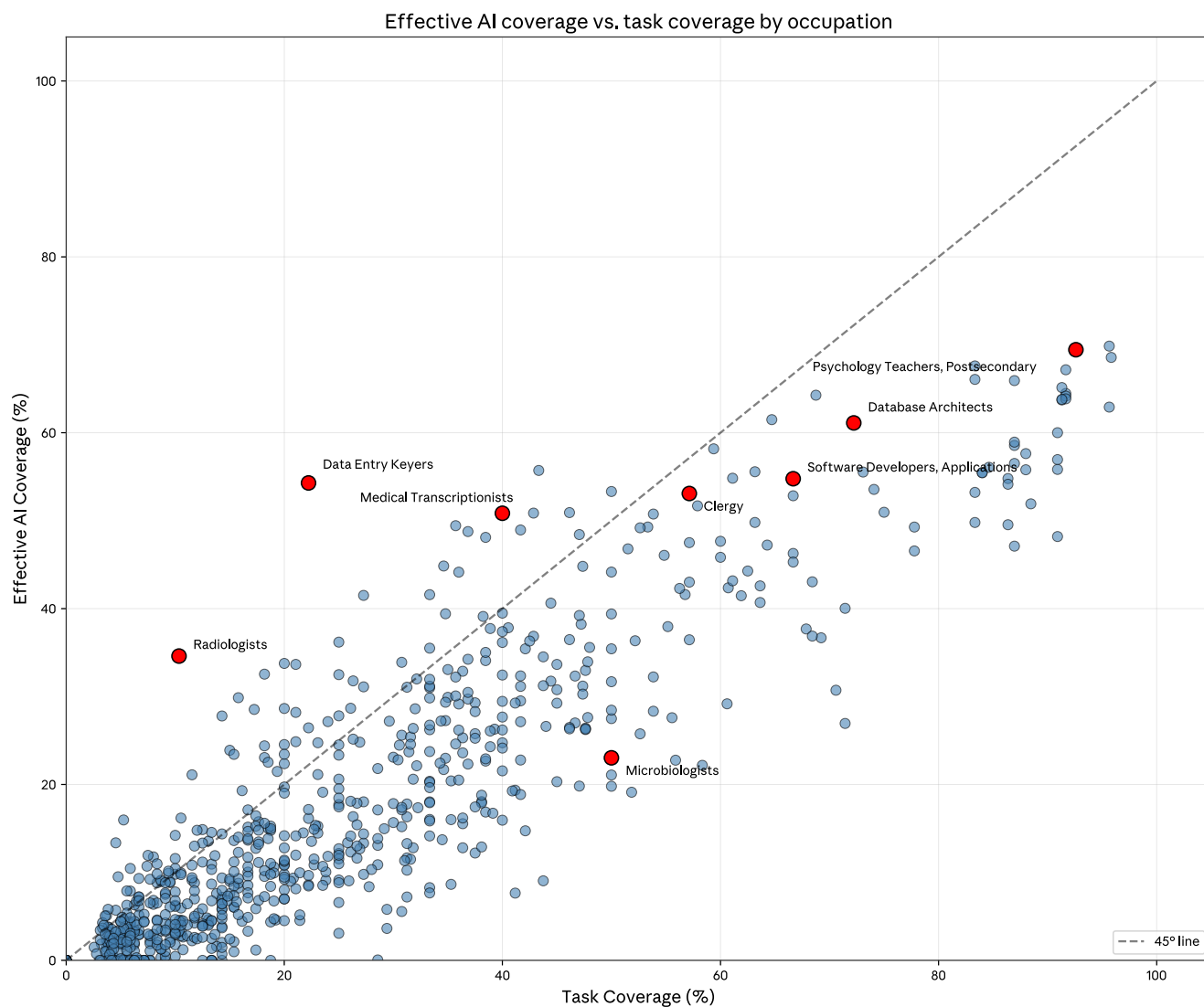


Figure 4.4: Effective AI coverage vs. Task coverage

The plot shows a scatter of the bivariate relationship between task effective AI coverage (%) and task coverage, measured at the occupation level. Effective AI coverage tracks the share of a worker's time-weighted duties that AI could successfully perform, based on Claude.ai data. Task coverage is the share of tasks that appear in Claude.ai usage. The dashed line shows where Effective AI coverage share equals task coverage.

Zooming in, several occupations show large differences in effective AI coverage compared to task coverage. For example, data entry workers have one of the highest effective AI coverage. This is because although only two of their nine tasks are covered, their largest task—reading and entering data from source documents—has high success rates with Claude. AI excels at what they spend most of their time doing.

Medical transcriptionists and radiologists also move up because their covered tasks happen to be their most time-intensive and highest-frequency work. For radiologists, their top two tasks—interpreting diagnostic images and